

# CROSS-EF-05 v1: CEweak and CL3 compatibility

ZFL Lab

September 3, 2026

State: C-PROVEN = {CEweak, CL3} in main. Tag checkpoint = {CEweak}.

CEweak :=  $\limsup \int_0^t \int \omega^T \text{Somega} = +infinity$  CL3 :=  $\limsup ||u||_{L3} = +infinity$

Mission: Check if CEweak AND CL3 compatible or contradiction.

No bot assumed. Expected: COMPATIBLE/OPEN.

Question: Does CL3 blowup imply CEweak blowup? Or independent? We test if  $\sup ||u||_{L3}$  finite would already imply CEweak finite? No theorem yet linking vortex stretching integral to L3 norm directly.

Verdict: COMPATIBLE, both necessary, independent.

Next: attempt inequality between  $\omega^T \text{Somega}$  and  $||u||_{L3}$ ?